# ImPRESS Model selection

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## Introduction

In this document we want to assess the model fit for different covariance structures in the primary statistical model used in ImPRESS. We use the Akaike Information Criterion (AIC) as the model selection criteria. The four models we want to compare are: unstructured, autoregressive and compound symmetry. Note that these analyses are based on sham randomisation, we deliberately ignore the true treatment allocation to avoid any information bias.

In addition we want to assess other diagnostic plots prior to unblining and final analyses to choose the correct models.

## Analyses for primary model fit

```r
admri <- tar_read(admri, store = "_targets")
tmp <- admri |>
    filter(paramcd == "mrt1cf" & cohortcd == 2) |>
    filter(avisitn <= 43) |>
    mutate(trt = case_when(
        avisitn == 15 & str_starts(step, "Cycle 1") ~ paste0(dose01p, " ", dose01pu),
        avisitn == 29 & str_starts(step, "Cycle 1") ~ paste0(dose01p, " ", dose01pu),
        avisitn == 43 & str_starts(step, "Cycle 1") ~ paste0(dose01p, " ", dose01pu),
        avisitn == 29 & str_starts(step, "Cycle 2") ~ paste0(dose01p, " ", dose01pu),
        avisitn == 43 & str_starts(step, "Cycle 2") ~ paste0(dose01p, " ", dose01pu),
        avisitn == 43 & str_starts(step, "Cycle 3") ~ paste0(dose01p, " ", dose01pu),
        TRUE ~ "0mg"
    )) |>
    mutate(
        trt = factor(trt, levels = c("0mg", "25 mg", "50 mg", "100 mg")),
        subjid = factor(subjid),
        avisitn = factor(avisitn, levels = c(-7, 15, 29, 43)
        )
    )

m <- mmrm(aval ~ trt + avisitn + us(avisitn | subjid),
    data = tmp
)



m2 <- mmrm(aval ~ trt + avisitn + ar1(avisitn | subjid),
    data = tmp
```

```r
)


m3 <- mmrm(aval ~ trt + avisitn + cs(avisitn | subjid),
    data = tmp
)
summary(m)
```

```
## mmrm fit
##
## Formula:     aval ~ trt + avisitn + us(avisitn | subjid)
## Data:        tmp (used 189 observations from 54 subjects with maximum 4
## timepoints)
## Covariance:  unstructured (10 variance parameters)
## Method:      Satterthwaite
## Vcov Method: Asymptotic
## Inference:   REML
##
## Model selection criteria:
##      AIC      BIC   logLik deviance
##    107.8    127.7    -43.9     87.8
##
## Coefficients:
##              Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)   1.48232    0.05661  53.00000  26.185   <2e-16 ***
## trt25 mg     -0.04666    0.06876 115.32000  -0.679   0.4988
## trt50 mg     -0.02259    0.06846 113.08000  -0.330   0.7420
## trt100 mg    -0.01511    0.06972 111.38000  -0.217   0.8288
## avisitn15     0.07852    0.03767  53.02000   2.085   0.0419 *
## avisitn29     0.06886    0.05273  71.51000   1.306   0.1958
## avisitn43    -0.02708    0.06923 121.19000  -0.391   0.6964
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Covariance estimate:
##        -7     15     29     43
## -7 0.1731 0.1738 0.1725 0.1509
## 15 0.1738 0.2261 0.2040 0.1817
## 29 0.1725 0.2040 0.2467 0.1951
## 43 0.1509 0.1817 0.1951 0.2345
```

```r
summary(m2)
```

```
## mmrm fit
##
## Formula:     aval ~ trt + avisitn + ar1(avisitn | subjid)
## Data:        tmp (used 189 observations from 54 subjects with maximum 4
## timepoints)
## Covariance:  auto-regressive order one (2 variance parameters)
## Method:      Satterthwaite
## Vcov Method: Asymptotic
## Inference:   REML
##
## Model selection criteria:
```

```
##      AIC      BIC   logLik deviance
##    106.9    110.9    -51.4    102.9
##
## Coefficients:
##              Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)   1.48232    0.06188  79.95000  23.954   <2e-16 ***
## trt25 mg     -0.05963    0.06936 137.14000  -0.860   0.3914
## trt50 mg     -0.01754    0.06967 138.52000  -0.252   0.8016
## trt100 mg    -0.04416    0.07126 138.72000  -0.620   0.5364
## avisitn15     0.08424    0.04065 124.68000   2.073   0.0403 *
## avisitn29     0.07786    0.05942 135.52000   1.310   0.1923
## avisitn43    -0.01776    0.07577 143.09000  -0.234   0.8150
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Covariance estimate:
##        -7     15     29     43
## -7 0.2068 0.1740 0.1463 0.1231
## 15 0.1740 0.2068 0.1740 0.1463
## 29 0.1463 0.1740 0.2068 0.1740
## 43 0.1231 0.1463 0.1740 0.2068
```

```
summary(m3)
```

```
## mmrm fit
##
## Formula:     aval ~ trt + avisitn + cs(avisitn | subjid)
## Data:        tmp (used 189 observations from 54 subjects with maximum 4
## timepoints)
## Covariance:  compound symmetry (2 variance parameters)
## Method:      Satterthwaite
## Vcov Method: Asymptotic
## Inference:   REML
##
## Model selection criteria:
##      AIC      BIC   logLik deviance
##    104.7    108.6    -50.3    100.7
##
## Coefficients:
##              Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)   1.48232    0.06343  69.45000  23.369   <2e-16 ***
## trt25 mg     -0.06180    0.07125 141.61000  -0.867    0.387
## trt50 mg      0.01137    0.07071 142.91000   0.161    0.872
## trt100 mg    -0.01273    0.07116 141.45000  -0.179    0.858
## avisitn15     0.07341    0.04575 132.10000   1.605    0.111
## avisitn29     0.06511    0.05608 134.33000   1.161    0.248
## avisitn43    -0.03178    0.06946 135.69000  -0.457    0.648
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Covariance estimate:
##        -7     15     29     43
## -7 0.2173 0.1770 0.1770 0.1770
## 15 0.1770 0.2173 0.1770 0.1770
## 29 0.1770 0.1770 0.2173 0.1770
```

```
## 43 0.1770 0.1770 0.1770 0.2173
```

```r
aic_table <- tibble(
    Model = c("Unstructured", "Autoregressive", "Compound Symmetry"),
    AIC = c(AIC(m), AIC(m2), AIC(m3))
)

aic_table
```

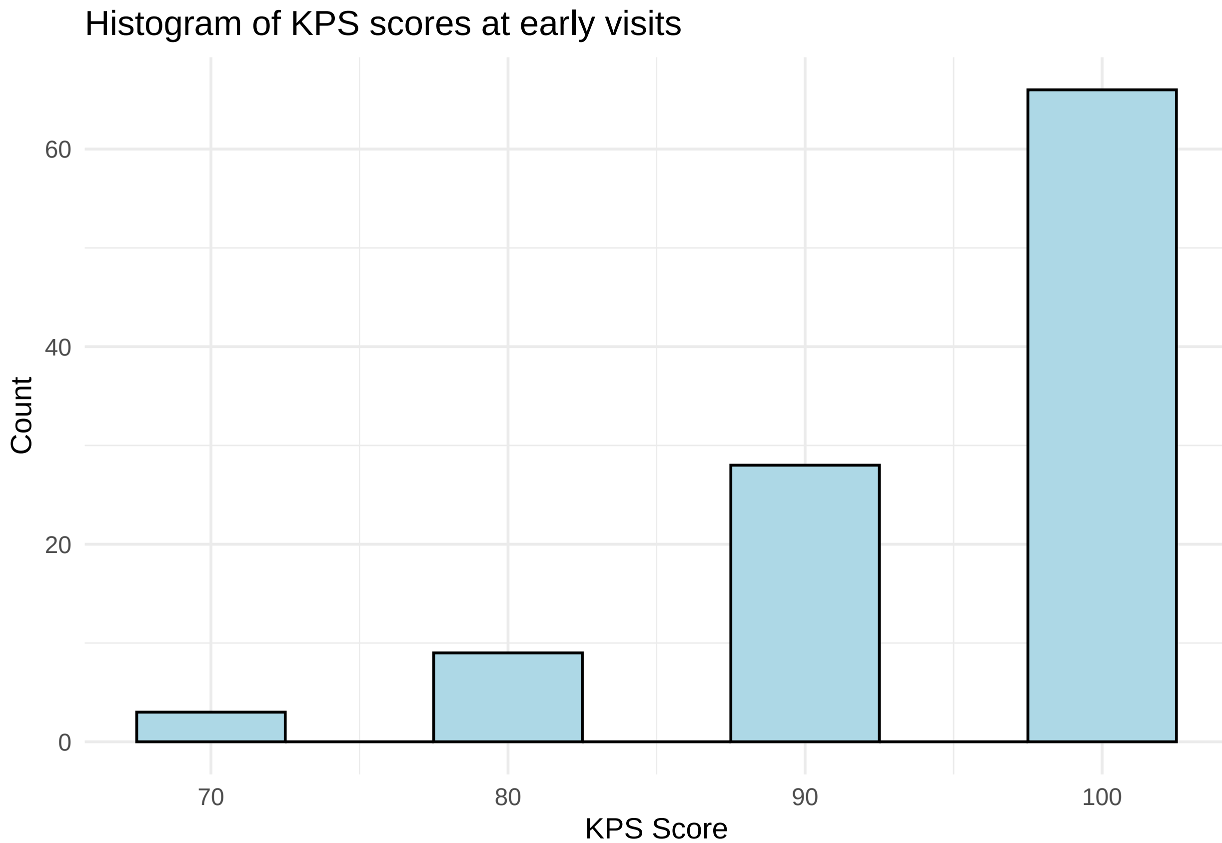
```
## # A tibble: 3 x 2
##   Model               AIC
##   <chr>             <dbl>
## 1 Unstructured       108.
## 2 Autoregressive     107.
## 3 Compound Symmetry  105.
```

### Conclusion on the primary model fit

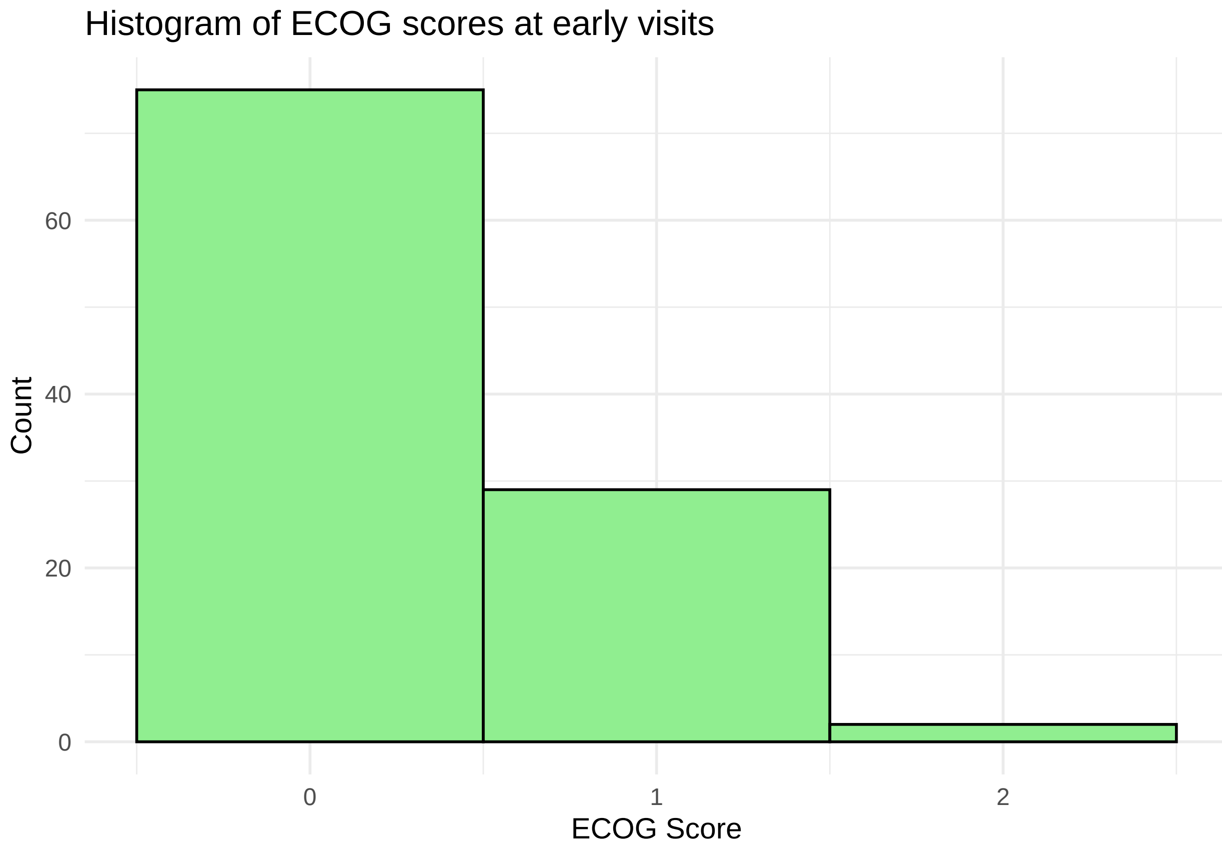
We see that the compound symmetry model gives the optimal fit according to AIC and will therefore be used in the primary analysis of ImPRESS. Note that we will use the corresponding restricted maximum likelihood estimates for the final model, and use mixed model with a random intercept for the modelling, but this is completely equivalent to the marginal model with compound symmetry covariance structure.

## Diagnostic plots for the neurological performance model

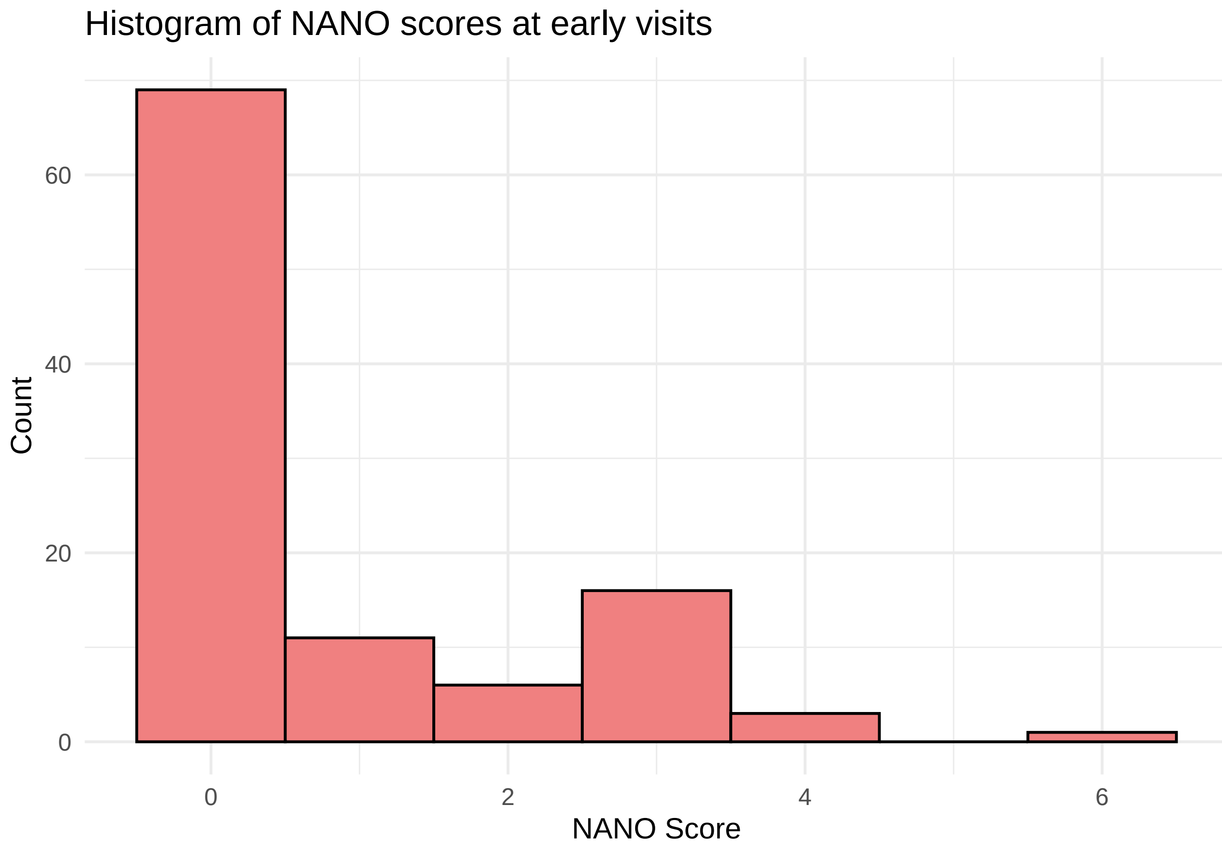
```r
adeff <- tar_read(adeff, store = "_targets")

# Make histograms of outcomes for kps, ecog and nano during the early visits

tmp_kps <- adeff |>
    filter(paramcd == "kps" & cohortcd == 2 & avisitn %in% c(15, 29, 43)) |>
    filter(cohortcd == 2)

tmp_ecog <- adeff |>
    filter(paramcd == "ecog" & cohortcd == 2 & avisitn %in% c(15, 29, 43)) |>
    filter(cohortcd == 2)

tmp_nano <- adeff |>
    filter(paramcd == "nano_tot" & cohortcd == 2 & avisitn %in% c(15, 29, 43)) |>
    filter(cohortcd == 2)

ggplot(tmp_kps, aes(x = aval)) +
    geom_histogram(binwidth = 5, fill = "lightblue", color = "black") +
    labs(title = "Histogram of KPS scores at early visits", x = "KPS Score", y = "Count") +
    theme_minimal()
```



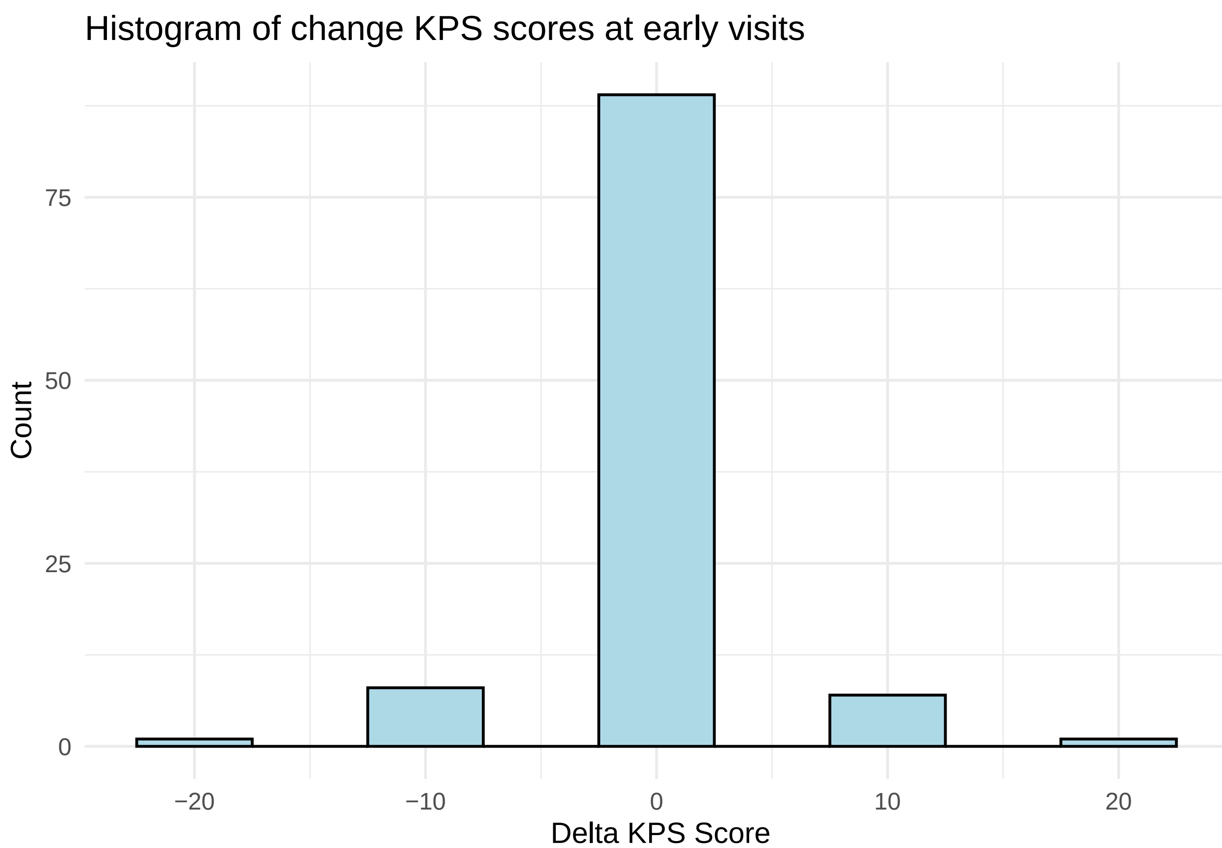

```r
ggplot(tmp_ecog, aes(x = aval)) +
    geom_histogram(binwidth = 1, fill = "lightgreen", color = "black") +
    labs(title = "Histogram of ECOG scores at early visits", x = "ECOG Score", y = "Count") +
    theme_minimal()
```



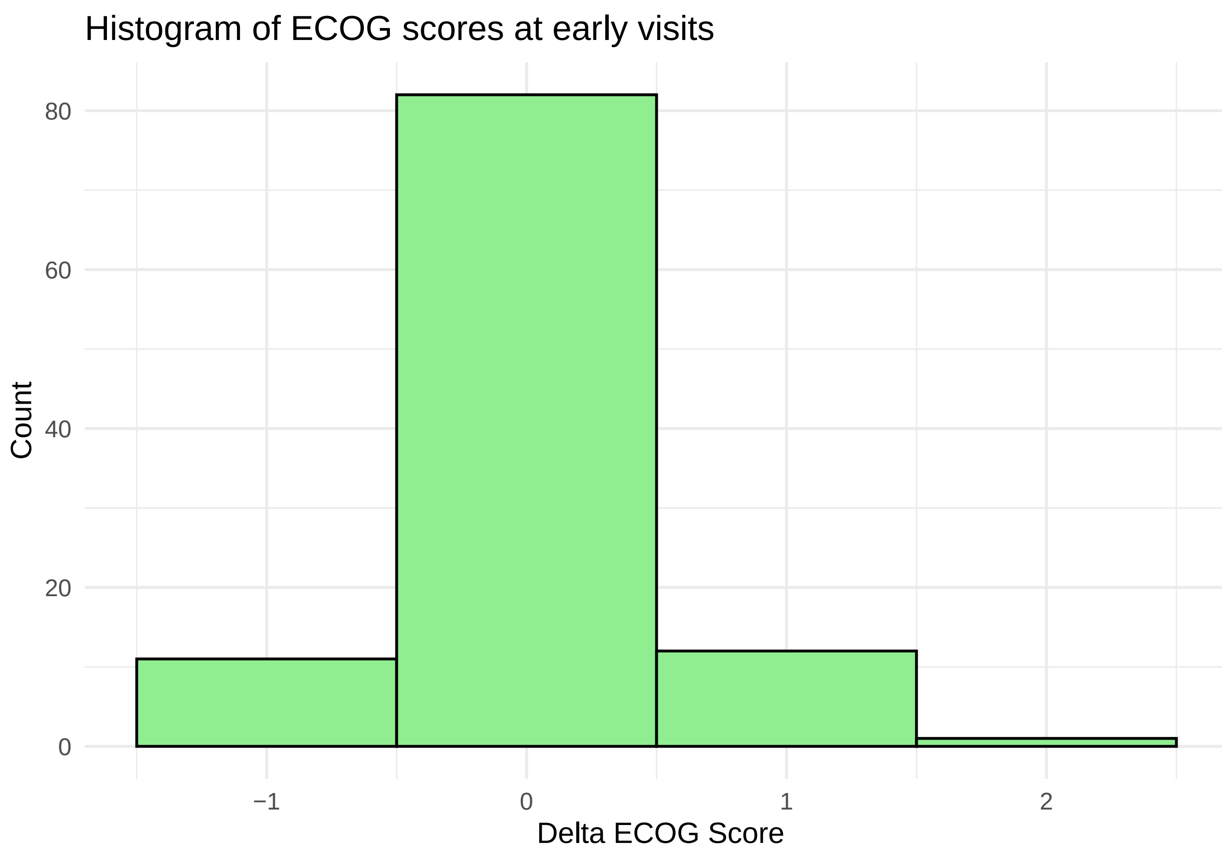

```r
ggplot(tmp_nano, aes(x = aval)) +
    geom_histogram(binwidth = 1, fill = "lightcoral", color = "black") +
    labs(title = "Histogram of NANO scores at early visits", x = "NANO Score", y = "Count") +
    theme_minimal()
```



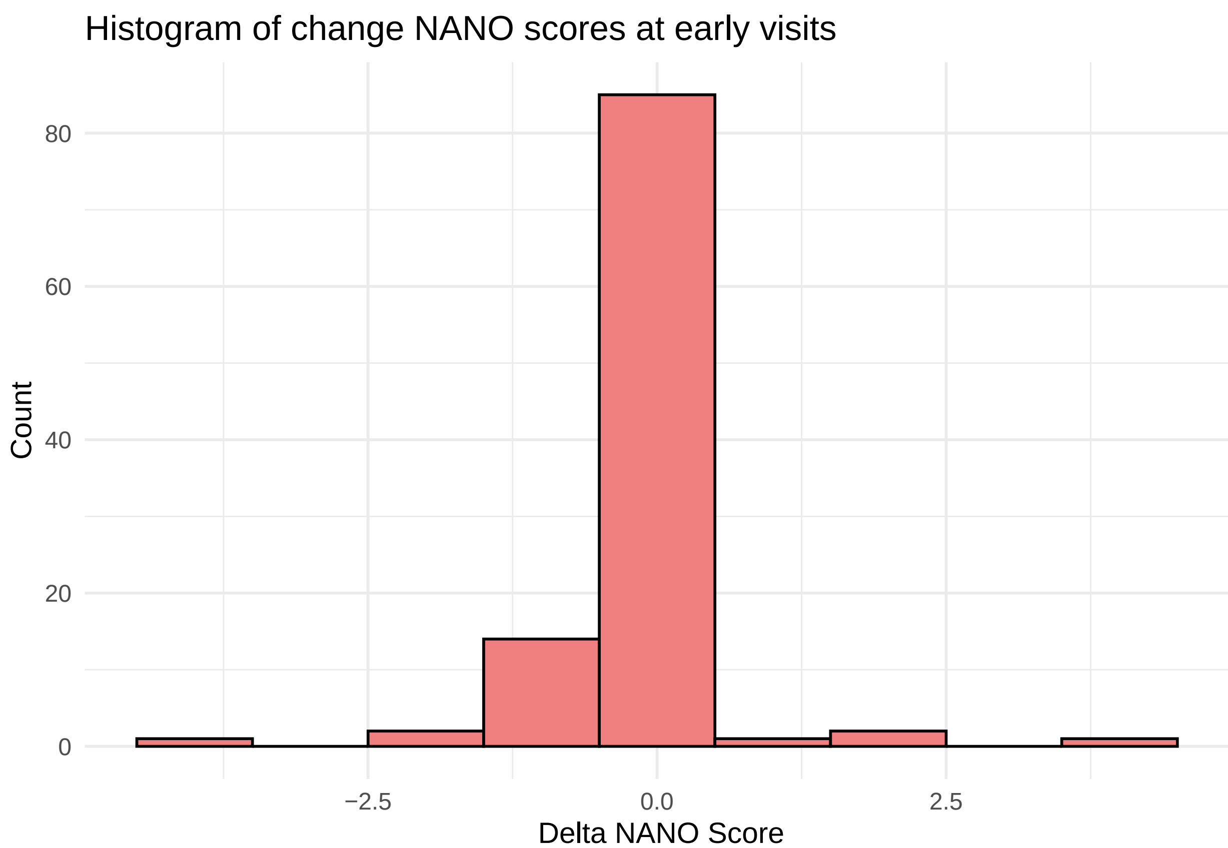

```
ggplot(tmp_kps, aes(x = chg)) +
    geom_histogram(binwidth = 5, fill = "lightblue", color = "black") +
    labs(title = "Histogram of change KPS scores at early visits", x = " Delta KPS Score", y = "Count")
    theme_minimal()
```




```r
ggplot(tmp_ecog, aes(x = chg)) +
    geom_histogram(binwidth = 1, fill = "lightgreen", color = "black") +
    labs(title = "Histogram of ECOG scores at early visits", x = " Delta ECOG Score", y = "Count") +
    theme_minimal()
```

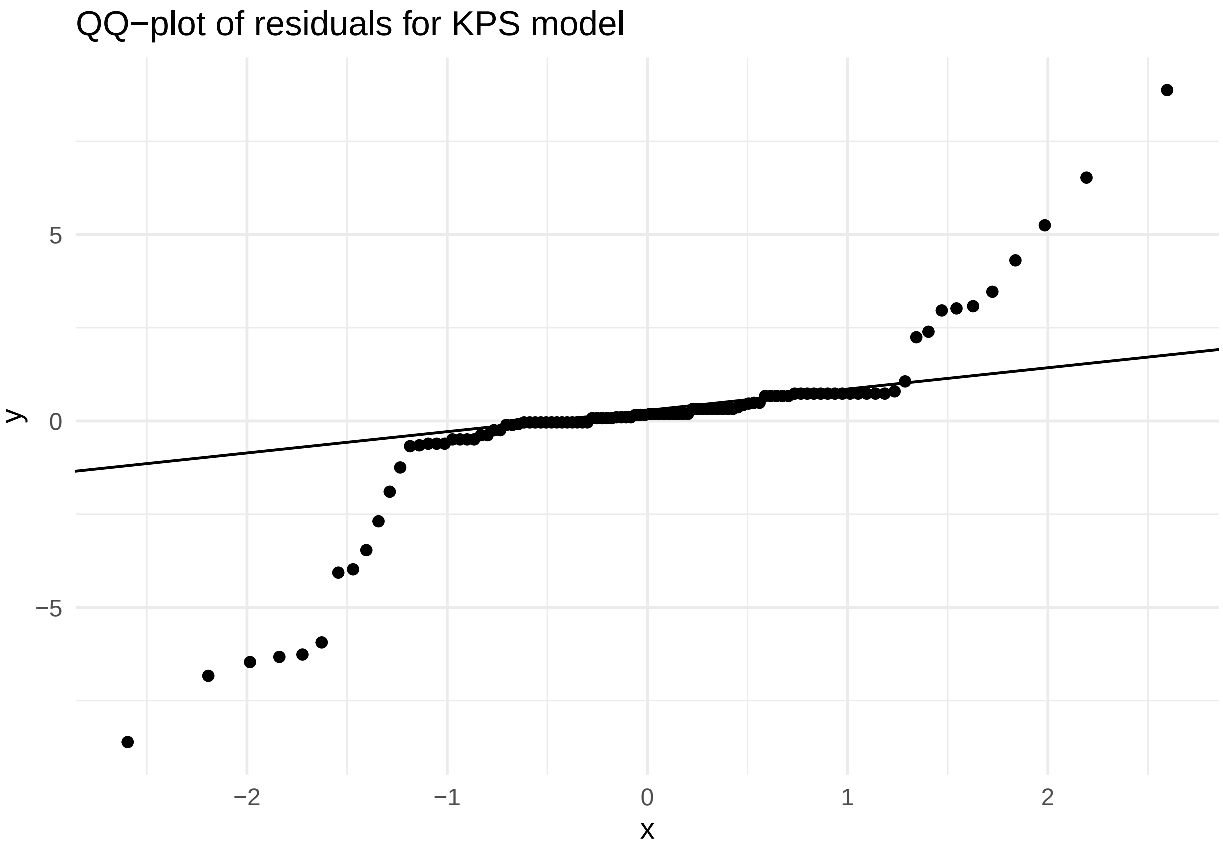



```
ggplot(tmp_nano, aes(x = chg)) +
    geom_histogram(binwidth = 1, fill = "lightcoral", color = "black") +
    labs(title = "Histogram of change NANO scores at early visits", x = " Delta NANO Score", y = "Count
    theme_minimal()
```

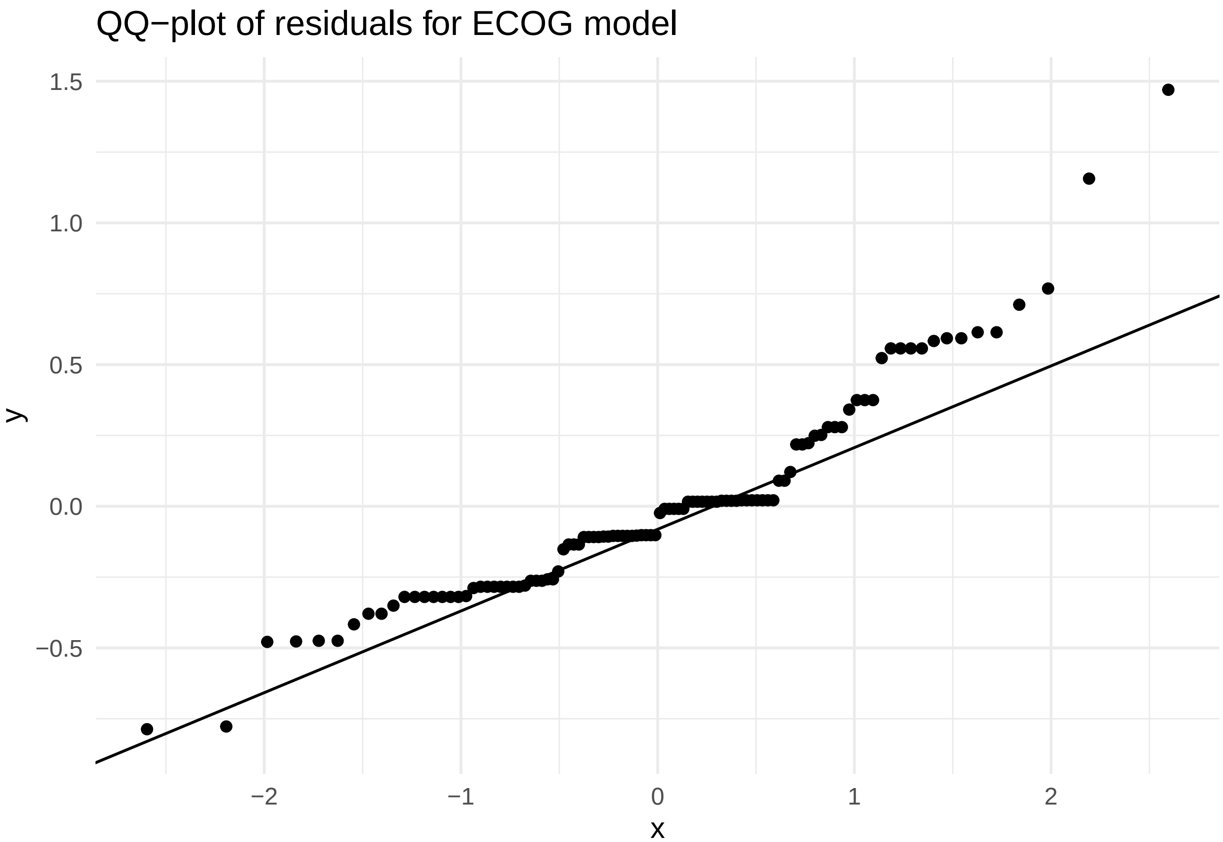



### Diagnostics for the residuals of the neurological performance models

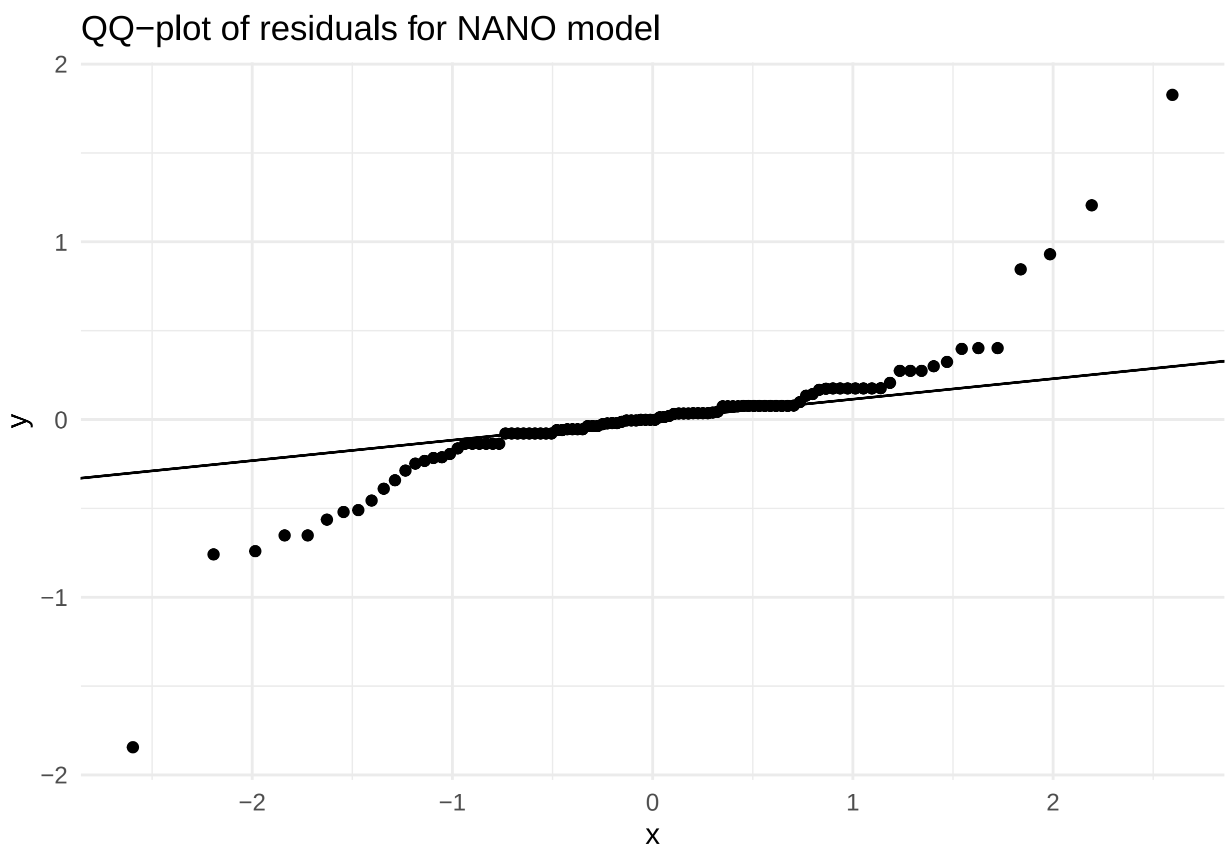
```
library(lme4)
  m_kps <- lme4::lmer(chg ~ base + trtcd + avisitn + (1 | usubjid), data = tmp_kps)
  # qq-plot of the residuals, Including a QQ line. Plot with ggplot2
  res <- resid(m_kps)
  ggplot(data = data.frame(res), aes(sample = res)) +
      stat_qq() +
      stat_qq_line() +
      labs(title = "QQ-plot of residuals for KPS model") +
      theme_minimal()
```




```r
m_ecog <- lme4::lmer(chg ~ base + trtcd + avisitn + (1 | usubjid), data = tmp_ecog)
# qq-plot of the residuals, Including a QQ line. Plot with ggplot2
res <- resid(m_ecog)
ggplot(data = data.frame(res), aes(sample = res)) +
    stat_qq() +
    stat_qq_line() +
    labs(title = "QQ-plot of residuals for ECOG model") +
    theme_minimal()
```




```r
m_nano <- lme4::lmer(chg ~ base + trtcd + avisitn + (1 | usubjid), data = tmp_nano)
# qq-plot of the residuals, Including a QQ line. Plot with ggplot2
res <- resid(m_nano)
ggplot(data = data.frame(res), aes(sample = res)) +
    stat_qq() +
    stat_qq_line() +
    labs(title = "QQ-plot of residuals for NANO model") +
    theme_minimal()
```

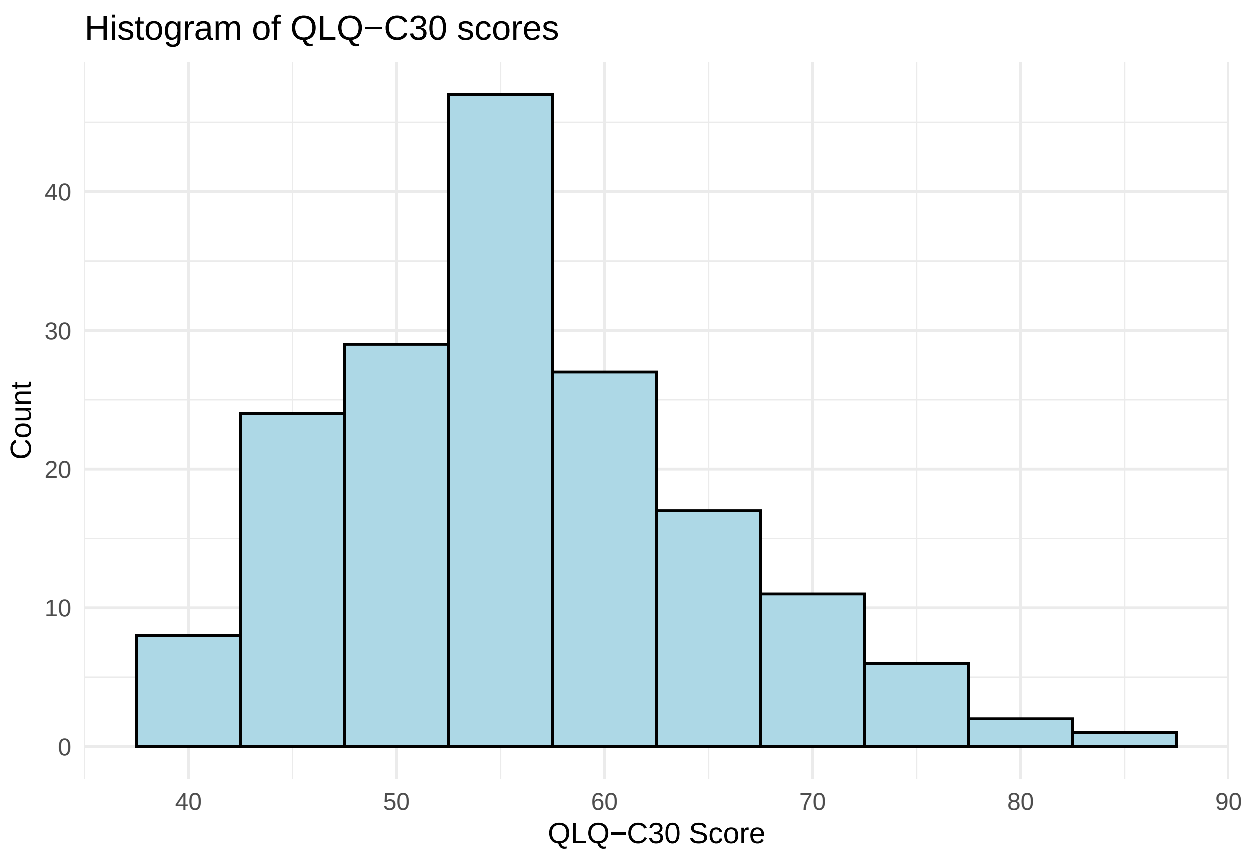



## Discussion on the histograms for neurological performance scores

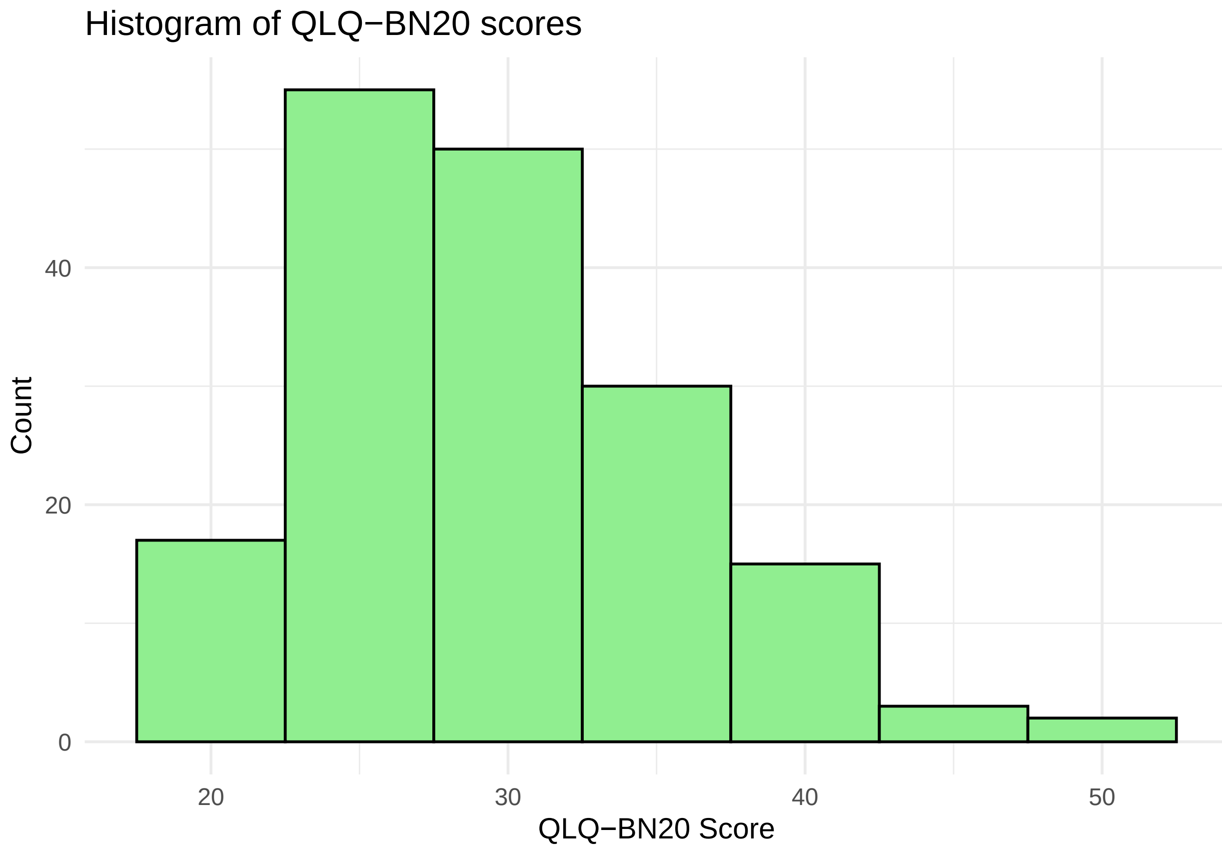
The histograms indicates very little change in the neurological performace scores. This is expected since the patients are newly diagnosed glioblastoma patients and the time frame is short. We therefore expect very limited treatment effects on these scores. These are very exploratory analyses, so we will keep the linear mixed regression model for these endpoints, noting the challenges in the interpretation of the results.

# Diagnostics for the Quality of Life fit_models

```
adeff <- tar_read(adeff, store = "_targets")

# Make histograms of outcomes for qlq_c30 and qlq_bn20

tmp_qlq_c30 <- adeff |>
    filter(paramcd == "qlq_c30" & cohortcd == 2) |>
    filter(ablfl != "Y")

tmp_qlq_bn20 <- adeff |>
    filter(paramcd == "qlq_bn20" & cohortcd == 2) |>
    filter(ablfl != "Y")


ggplot(tmp_qlq_c30, aes(x = aval)) +
    geom_histogram(binwidth = 5, fill = "lightblue", color = "black") +
    labs(title = "Histogram of QLQ-C30 scores", x = "QLQ-C30 Score", y = "Count") +
    theme_minimal()
```

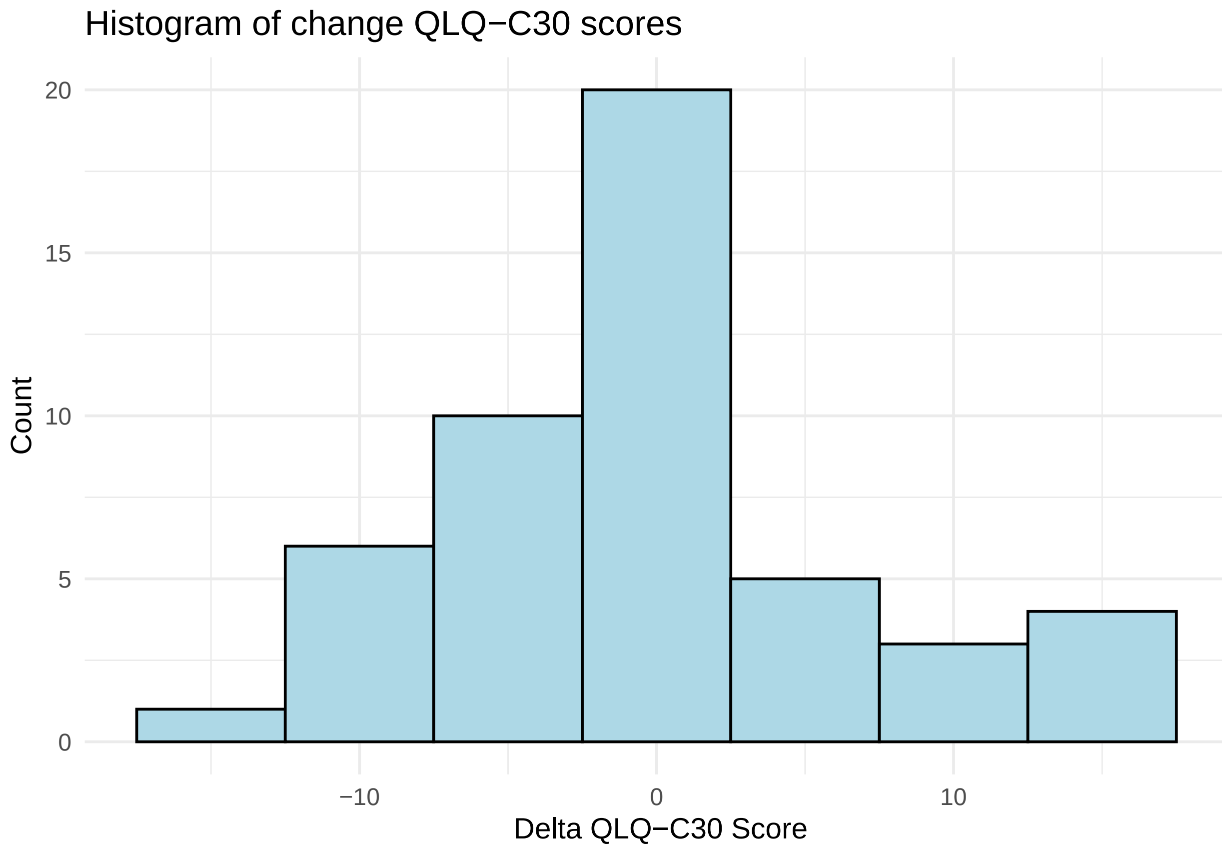



```r
ggplot(tmp_qlq_bn20, aes(x = aval)) +
    geom_histogram(binwidth = 5, fill = "lightgreen", color = "black") +
    labs(title = "Histogram of QLQ-BN20 scores", x = "QLQ-BN20 Score", y = "Count") +
    theme_minimal()
```



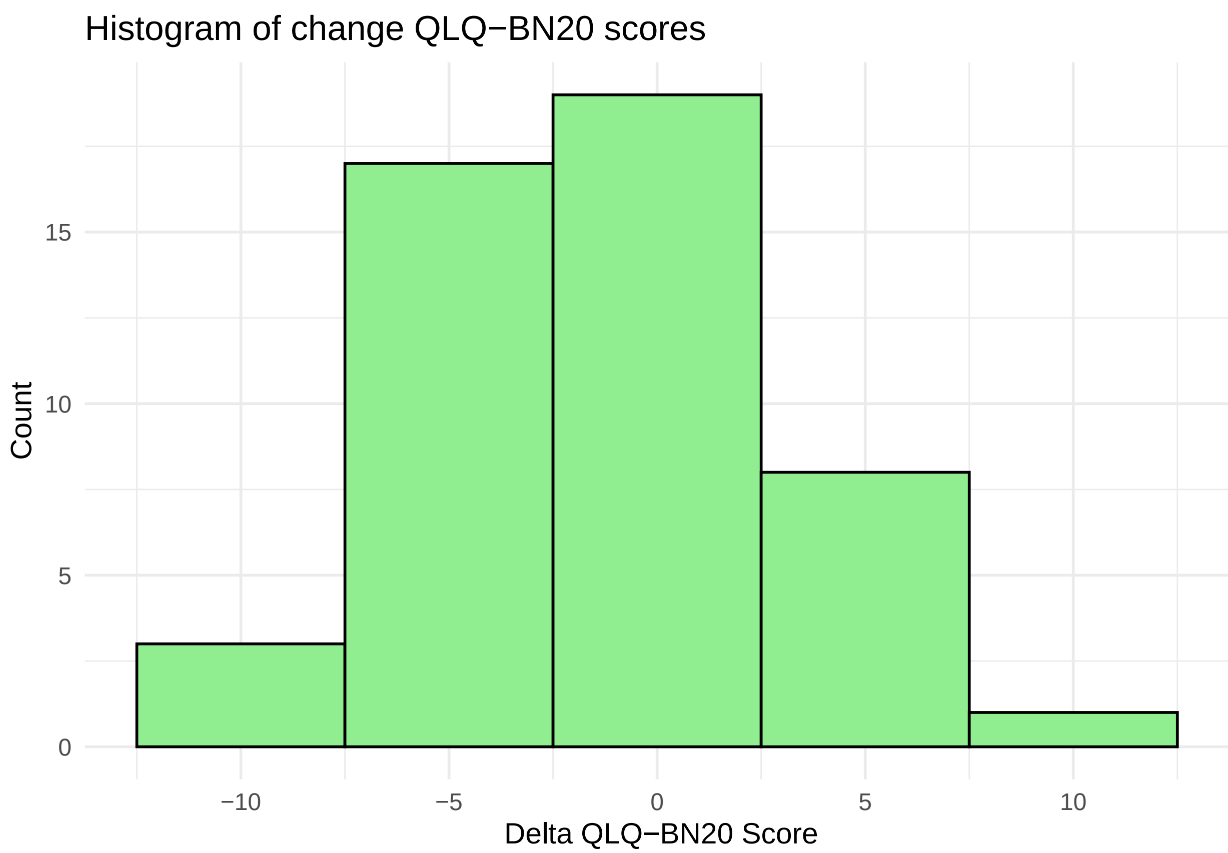

```r
# and then change scores

ggplot(tmp_qlq_c30, aes(x = chg)) +
    geom_histogram(binwidth = 5, fill = "lightblue", color = "black") +
    labs(title = "Histogram of change QLQ-C30 scores", x = " Delta QLQ-C30 Score", y = "Count") +
    theme_minimal()
```

```
## Warning: Removed 123 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
ggplot(tmp_qlq_bn20, aes(x = chg)) +
    geom_histogram(binwidth = 5, fill = "lightgreen", color = "black") +
    labs(title = "Histogram of change QLQ-BN20 scores", x = " Delta QLQ-BN20 Score", y = "Count") +
    theme_minimal()
```

```
## Warning: Removed 124 rows containing non-finite outside the scale range
## (`stat_bin()`).
```


Histogram of change QLQ−BN20 scores
Count
15
10
5
0
−10
−5
0
5
10
Delta QLQ−BN20 Score